

# **ME 447.01**

# **ENGINEERING FLUID**

# **DYNAMICS**

Aerodynamic Performance Analysis of  
NACA 2412 Airfoil Using Simflow

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## **1-Introduction**

The objective of this project is to investigate the aerodynamic performance of the NACA 2412 airfoil using Computational Fluid Dynamics (CFD) methods. According to the NACA 4-digit nomenclature, for the "2412" profile: the first digit "2" indicates that the maximum camber is 2% of the chord length, the second digit "4" indicates that this maximum camber is located at 40% of the chord from the leading edge, and the last two digits "12" indicate a maximum thickness of 12% of the chord.

In this study, the flow around the airfoil was modeled at a Reynolds number of  $Re = 150,000$  using SimFlow software. Analyses were performed at angles of attack (AoA) of  $0^\circ$ ,  $3^\circ$ ,  $6^\circ$ ,  $9^\circ$  and a higher angle where stall was clearly observed. The Lift Coefficient ( $C_L$ ), Drag Coefficient ( $C_D$ ), and Lift-to-Drag ratio ( $C_L/C_D$ ) were obtained and evaluated.

## 2-Airfoil Geometry NACA 2412

The airfoil geometry was generated by obtaining the NACA 2412 coordinates from the Airfoil Tools database in .dat format and importing them into the SimFlow workspace using the "Airfoil Mesh" module.

### Solver Settings:

**Flow Type:** Incompressible, Steady-State.

**Turbulence Model:** The k- $\omega$  SST model, which is the industry standard for aerospace and boundary layer analyses, was utilized.

**Solver:** The SIMPLE algorithm was chosen, and residual tolerances were tightened to  $10^{-6}$  to ensure excellent precision.

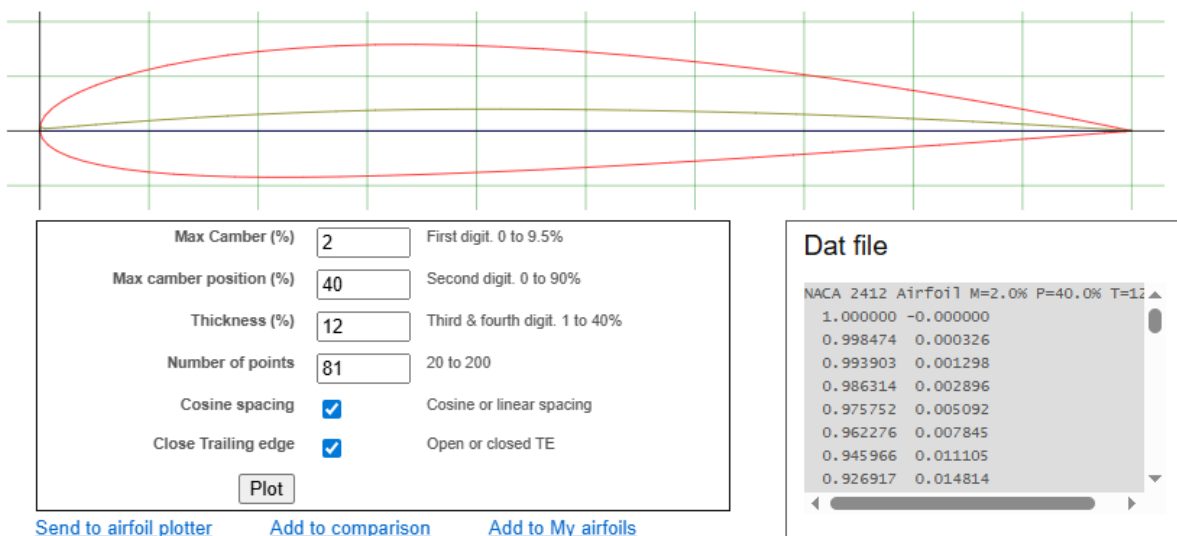
**Velocity Calculation:** The freestream velocity at the inlet boundary condition was defined to satisfy the  $Re=150,000$  requirement, based on the fluid's kinematic viscosity and the chord length.



Define New Parameter

Name  Formula

Figure 1: Definition of freestream velocity parameter for  $Re = 150,000$



Max Camber (%)  First digit. 0 to 9.5%

Max camber position (%)  Second digit. 0 to 90%

Thickness (%)  Third & fourth digit. 1 to 40%

Number of points  20 to 200

Cosine spacing ☒ Cosine or linear spacing

Close Trailing edge ☒ Open or closed TE

[Send to airfoil plotter](#) [Add to comparison](#) [Add to My airfoils](#)

Dat file

```
NACA 2412 Airfoil M=2.0% P=40.0% T=12
1.000000 -0.000000
0.998474 0.000326
0.993903 0.001298
0.986314 0.002896
0.975752 0.005092
0.962276 0.007845
0.945966 0.011105
0.926917 0.014814
```

Figure 2: Properties and image of NACA 2412

| Parameter       | Value |
|-----------------|-------|
| Maximum Camber  | %2    |
| Camber Position | %40   |
| Thickness       | %12   |

### 3- CFD Setup

| Parameter        | Value               |
|------------------|---------------------|
| Solver           | OpenFOAM (SimFlow)  |
| Flow Type        | Steady-State        |
| Fluid            | Air                 |
| Turbulence Model | k- $\omega$ SST     |
| AoA              | 0°, 3°, 6°, 9°, 15° |
| Airfoil          | NACA 2412           |

AIRFOIL MESH

Airfoil

2412.dat

Geometry

Radius [m]

25

Tail Length Ratio [-]

1.5

Tail Angle [deg]

0

Angle of Attack [deg]

0

Mesh

Surface Cell Thickness [m]

2e-04

Min Surface Cell Length [m]

2e-03

Max Surface Cell Length [m]

8e-03

Grading [-]

1.1

Cell Count

60059

Restore Previous Mesh

Mesh

SolversSIMPLEResidualsRelaxationLimits

pUk $\omega$

Solver

PBiCG Stabilized

Preconditioner

DILU

Max Iterations

100

+

-

Tolerance

1e-07

Relative Tolerance

1e-02

Figure 3 and 4: Airfoil mesh generation parameters and velocity (U) solver settings.

Solvers
SIMPLE
Residuals
Relaxation
Limits

p
U
k
 $\omega$

Solver

GAMG

Smoother

Gauss-Seidel

Cache Agglomeration

☒

Agglomerator

Face Area Pair

Cells In Coarsest Level

10

+

-

Merge Levels

1

+

-

Pre Sweeps

0

+

-

Post Sweeps

1

+

-

Finest Sweeps

2

+

-

Max Iterations

100

+

-

Tolerance

1e-07

Relative Tolerance

1e-02

Figure 5: Pressure (p) solver configuration utilizing GAMG

## 4-Results and Discussion

The steady-state aerodynamic coefficients obtained at the end of the iterations for different angles of attack are summarized in the table below:

| Angle of Attack (deg) | Lift Coefficient (CL) | Drag Coefficient (CD) | Efficiency (CL/CD) |
|-----------------------|-----------------------|-----------------------|--------------------|
| 0                     | 0.228                 | 0.0089                | 25.61              |
| 3                     | 0.543                 | 0.0105                | 51.71              |
| 6                     | 0.837                 | 0.0138                | 60.65              |
| 9                     | 1.150                 | 0.0187                | 61.49              |
| 15                    | 1.010                 | 0.0303                | 33.33              |

Figure 6: Aerodynamic Coefficient Table

## 4.1 Aerodynamic Coefficients (Graphs)

### Cl vs AoA (0, 3, 6, 9, 15)

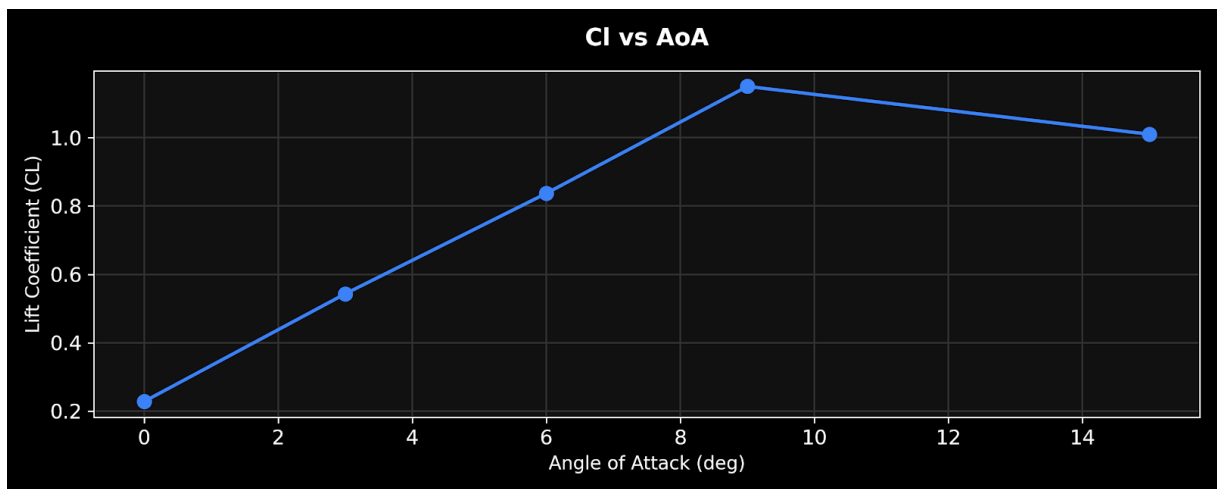


Figure 7: Cl vs AoA

### Cd vs. AoA (0, 3, 6, 9, 15)

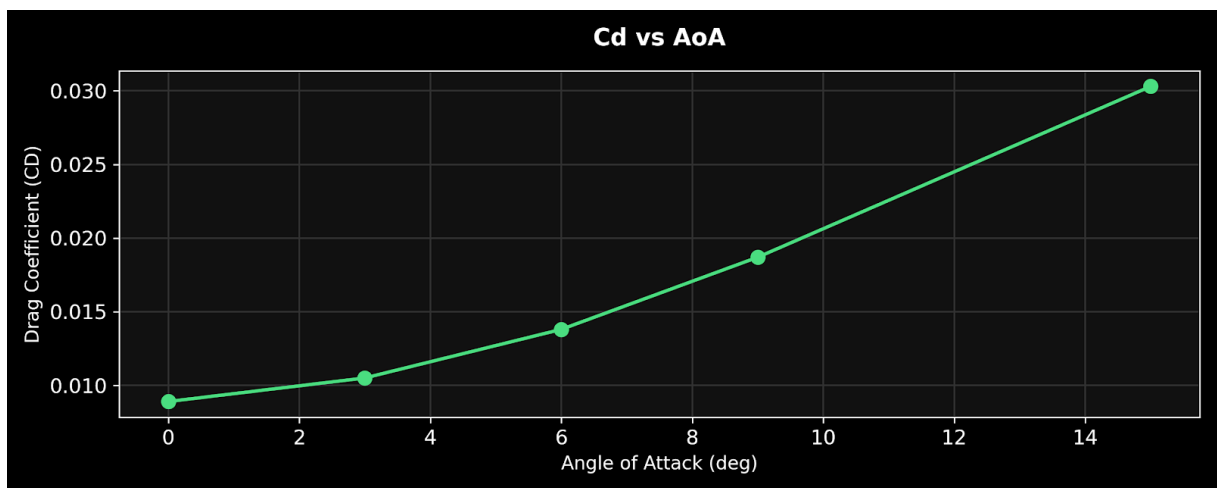


Figure 8: Cd vs AoA

### Cl vs. Cd (0, 3, 6, 9, 15)

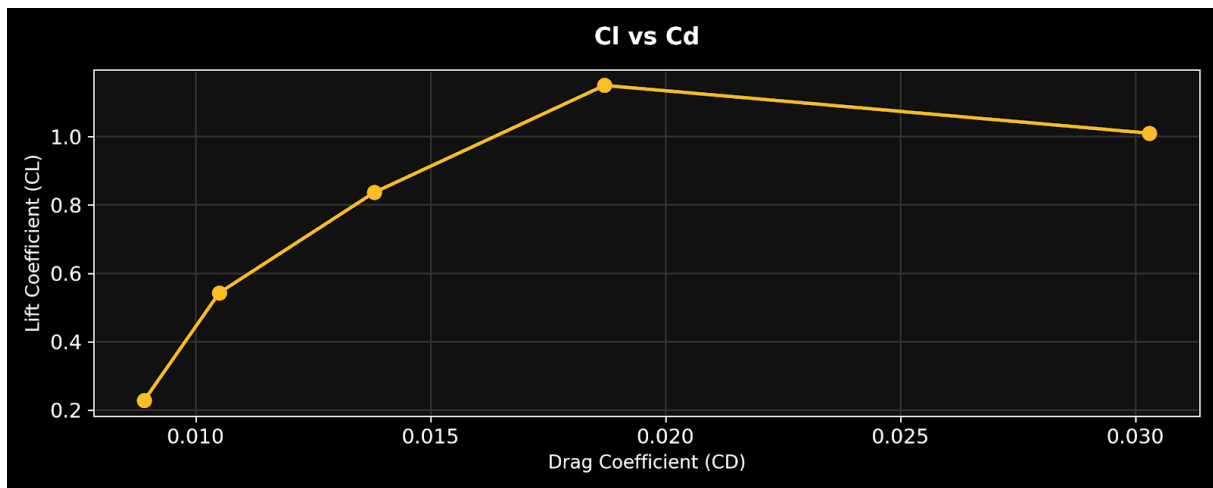


Figure 9: Cl vs Cd

### Cl/Cd vs. AoA (0, 3, 6, 9, 15)

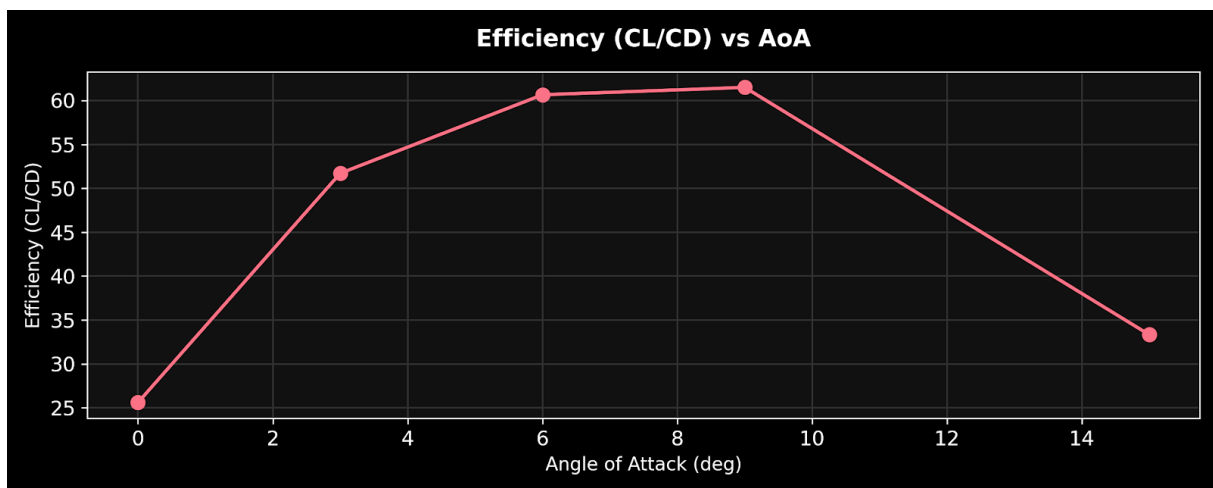


Figure 10: Cl/Cd vs AoA

**Interpretation of Graphs:** Examining the lift coefficient ( $C_L$ ) graph, it is observed that the lift force increases linearly as the angle of attack increases from  $0^\circ$  to  $9^\circ$ . At a  $9^\circ$  angle of attack, the airfoil reaches its maximum lift potential with ( $C_L$ ) = 1.150. However, increasing the angle beyond this point causes the ( $C_L$ ) value to drop to 1.010 and the drag coefficient ( $C_D$ ) to spike parabolically to 0.0303. This condition proves that the critical stall angle has been exceeded. The highest aerodynamic efficiency ( $C_L/C_D$ ) was achieved around  $9^\circ$  (with a ratio of 61.49).

## 4.2 Velocity and Pressure Distributions

Velocity (u-vel) and pressure contour maps for the analyzed angles are presented below.

Results of 0 Degrees AoA:

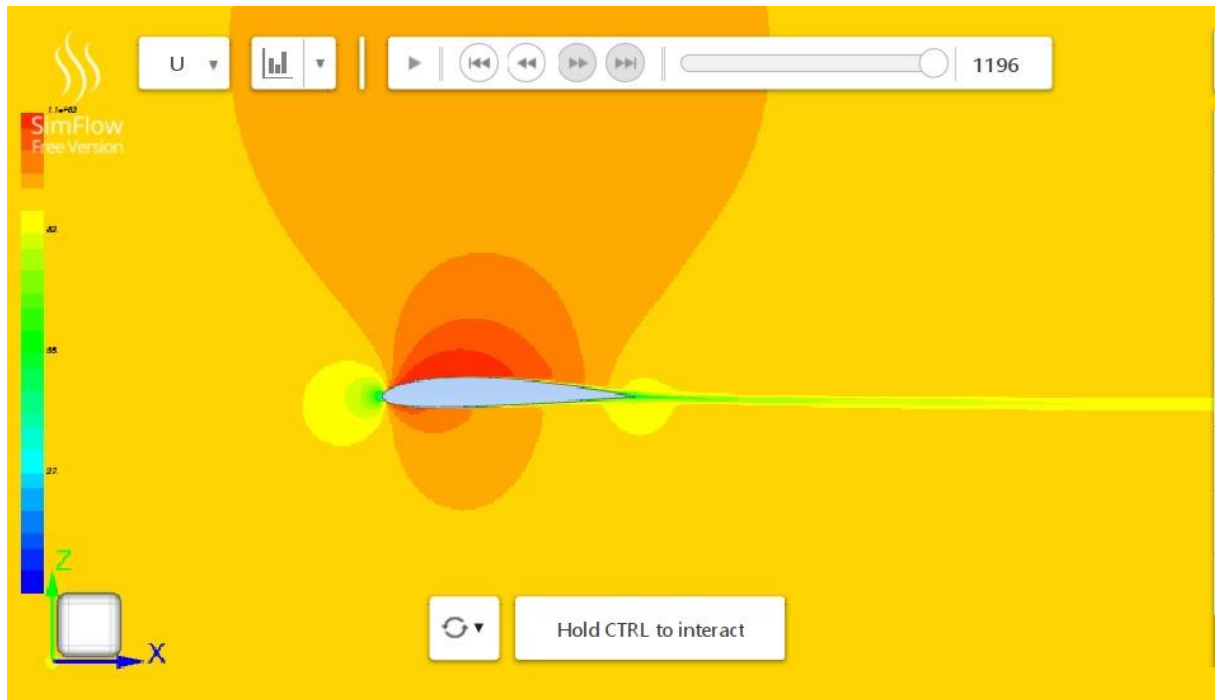


Figure 11: Velocity Contour

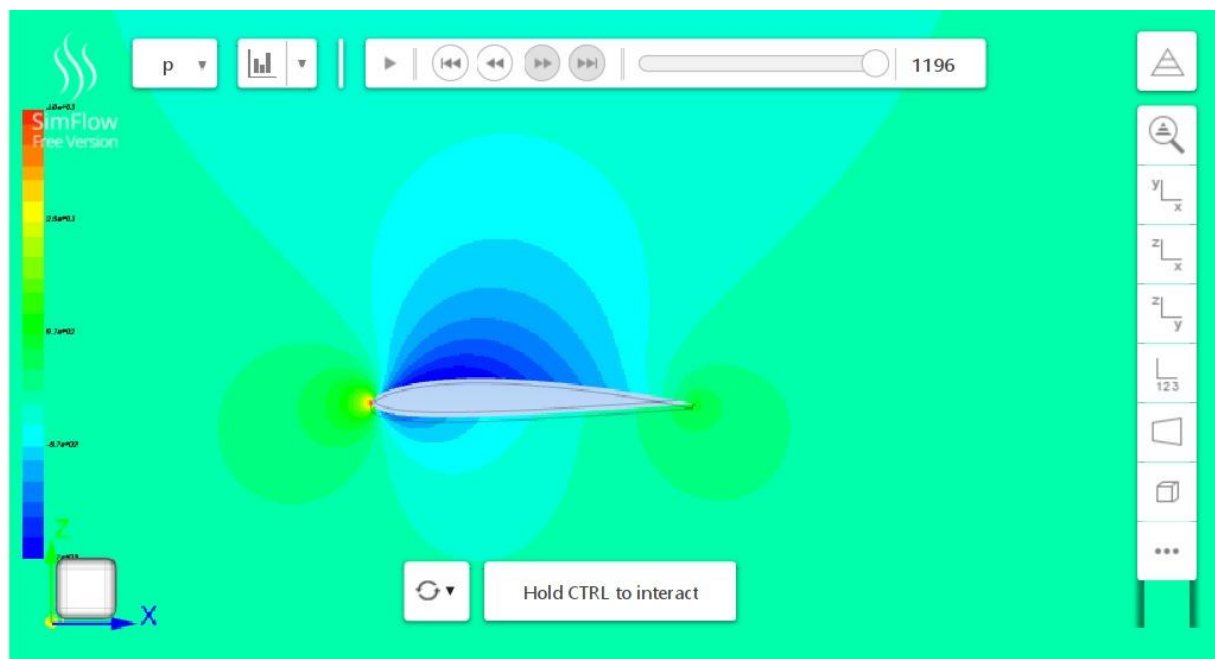


Figure 12: Pressure Contour

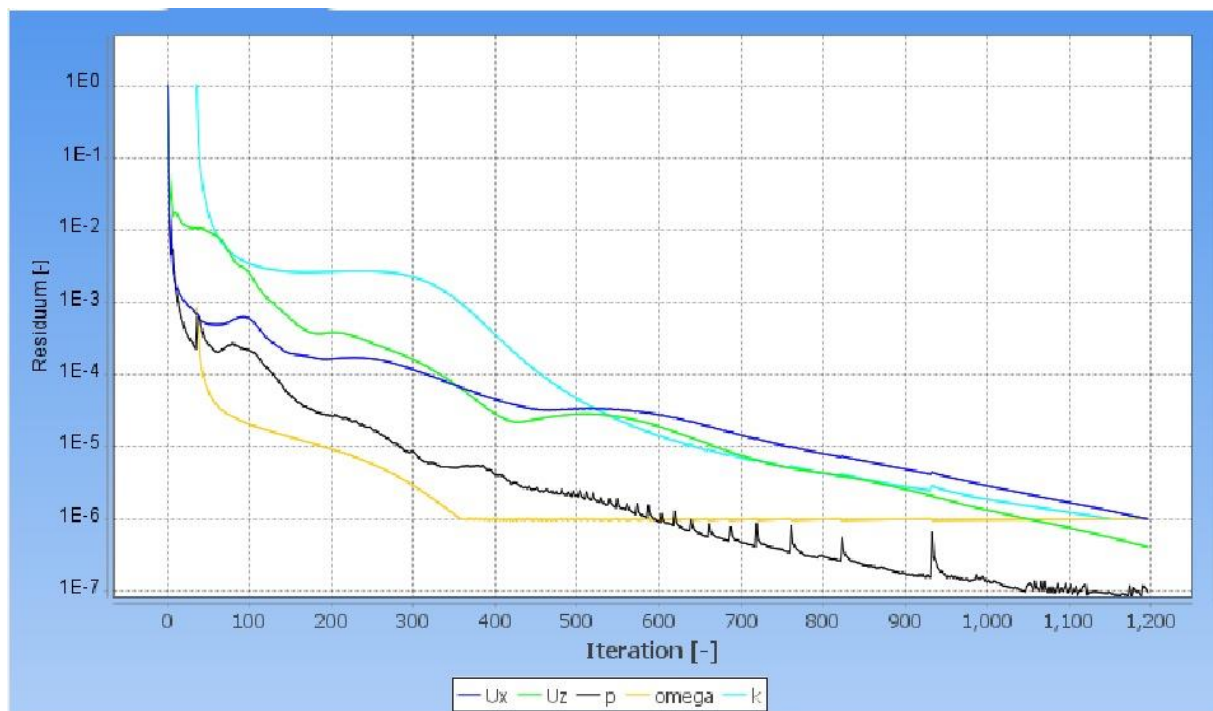


Figure 13: Residuals

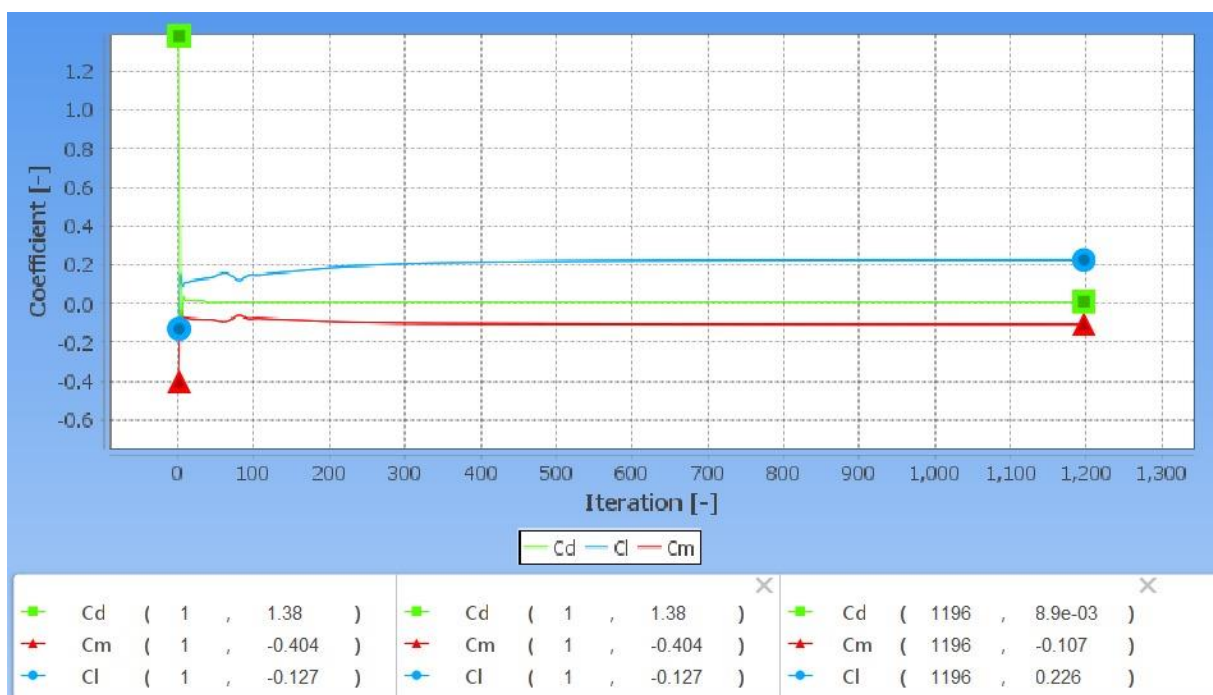


Figure 14:  $(C_d)$  ,  $(C_l)$ ,  $(C_m)$  values for 0-degree angle of attack

Results of 3 Degrees AoA:

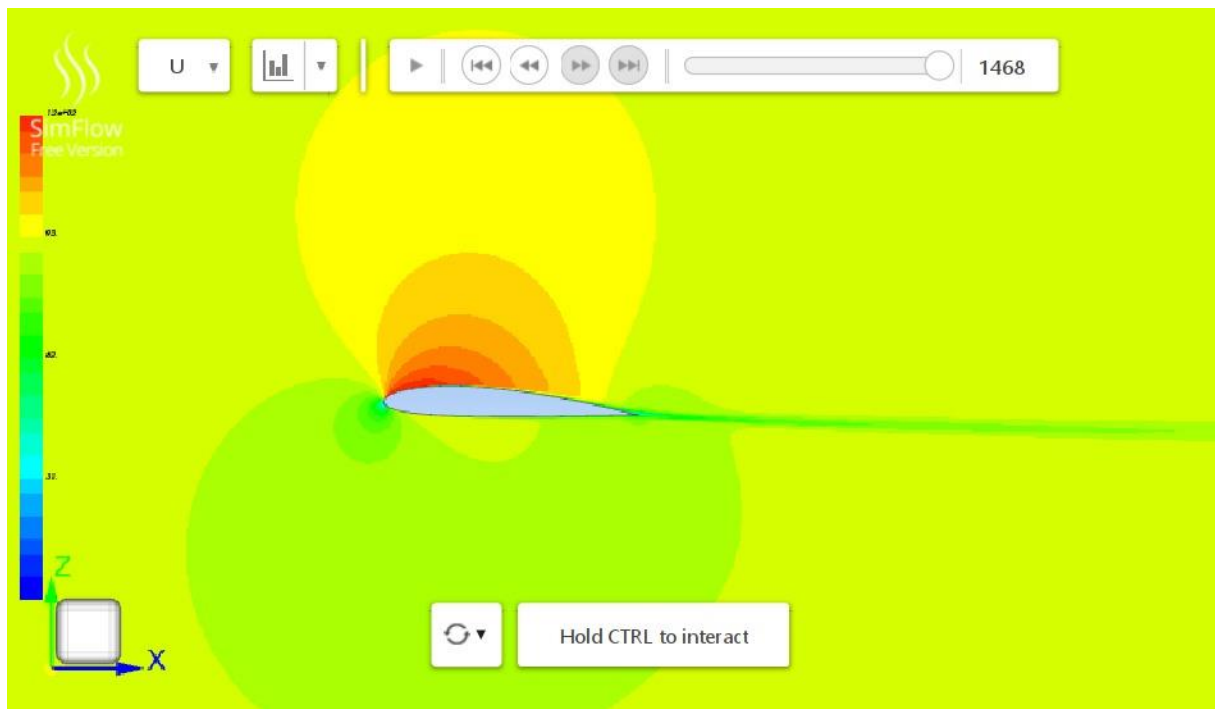


Figure 15: Velocity Contour

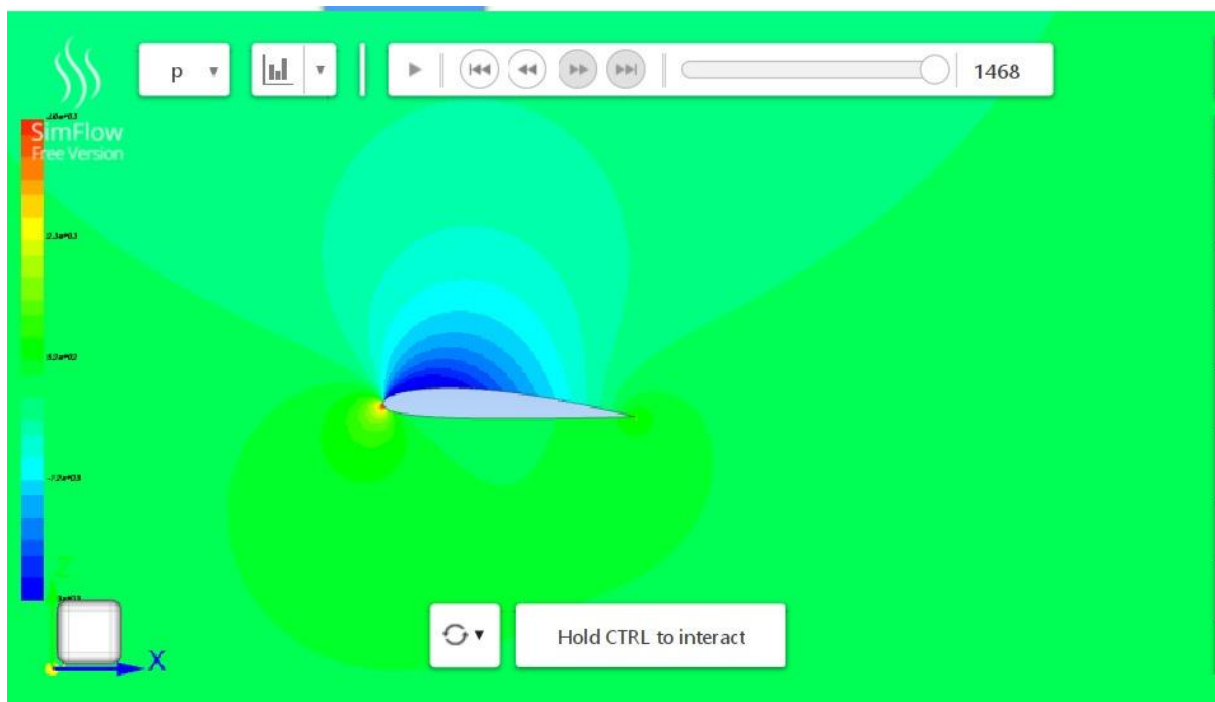


Figure 16: Pressure Contour

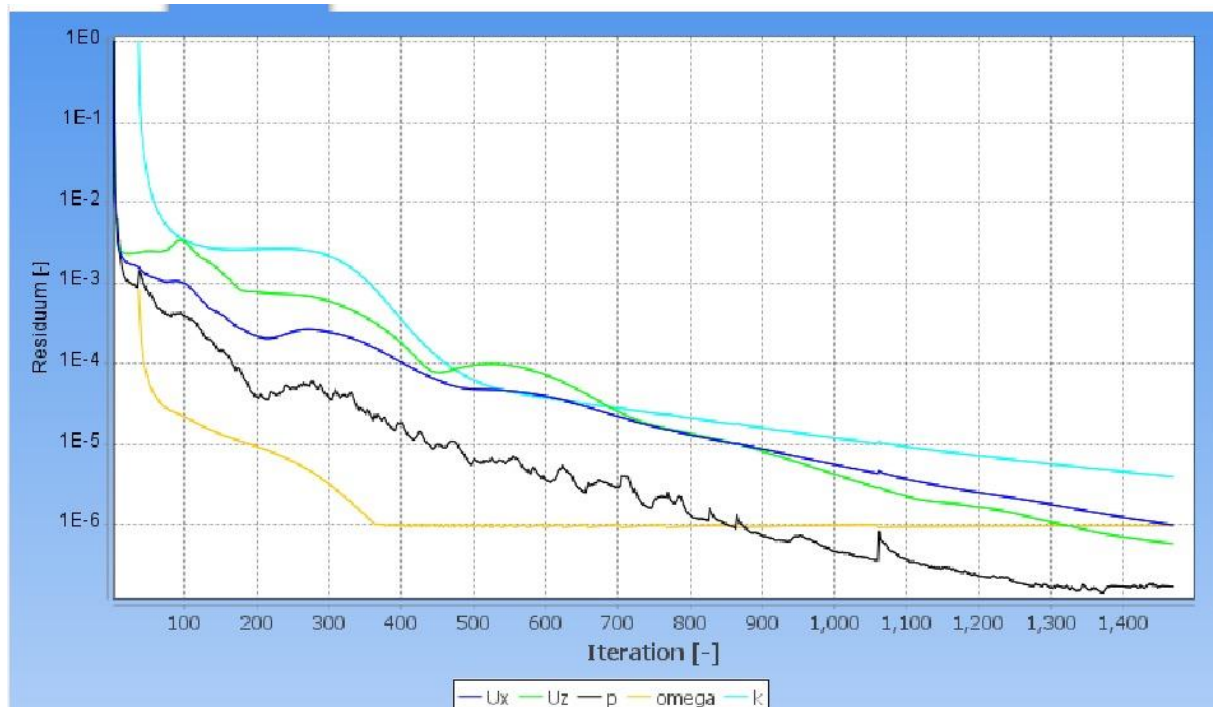


Figure 17: Residuals

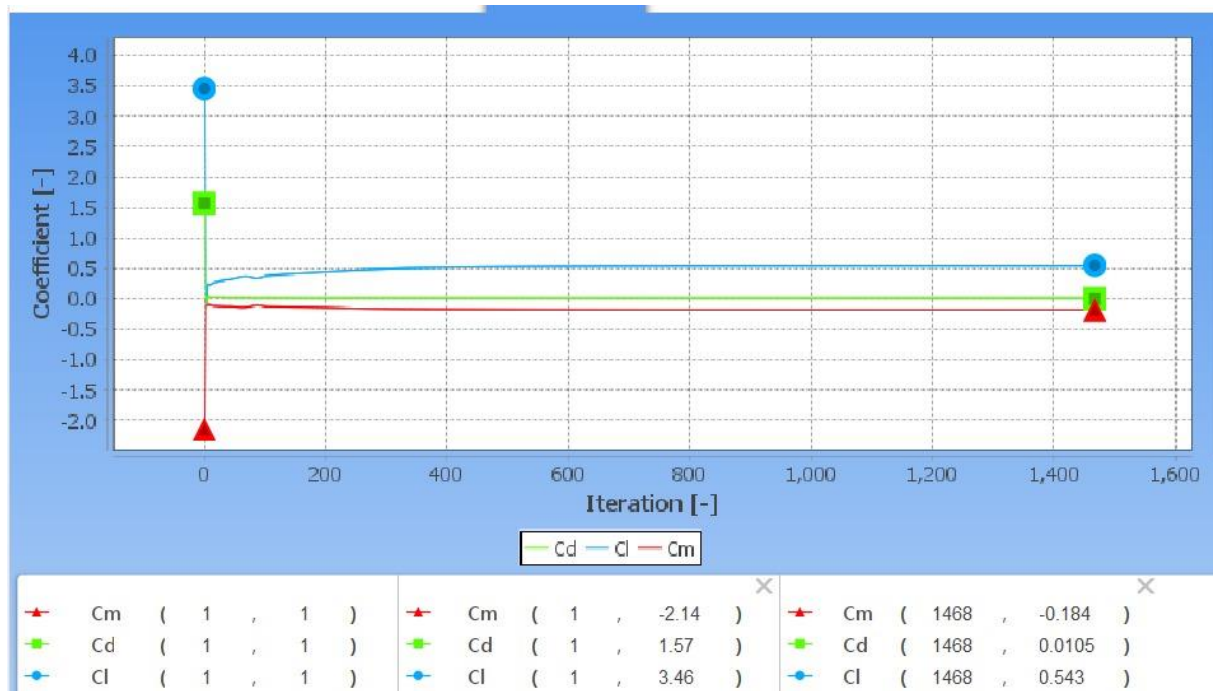


Figure 18:  $(C_d)$ ,  $(C_l)$ ,  $(C_m)$  values for 3-degree angle of attack

Results of 6 Degrees AoA:

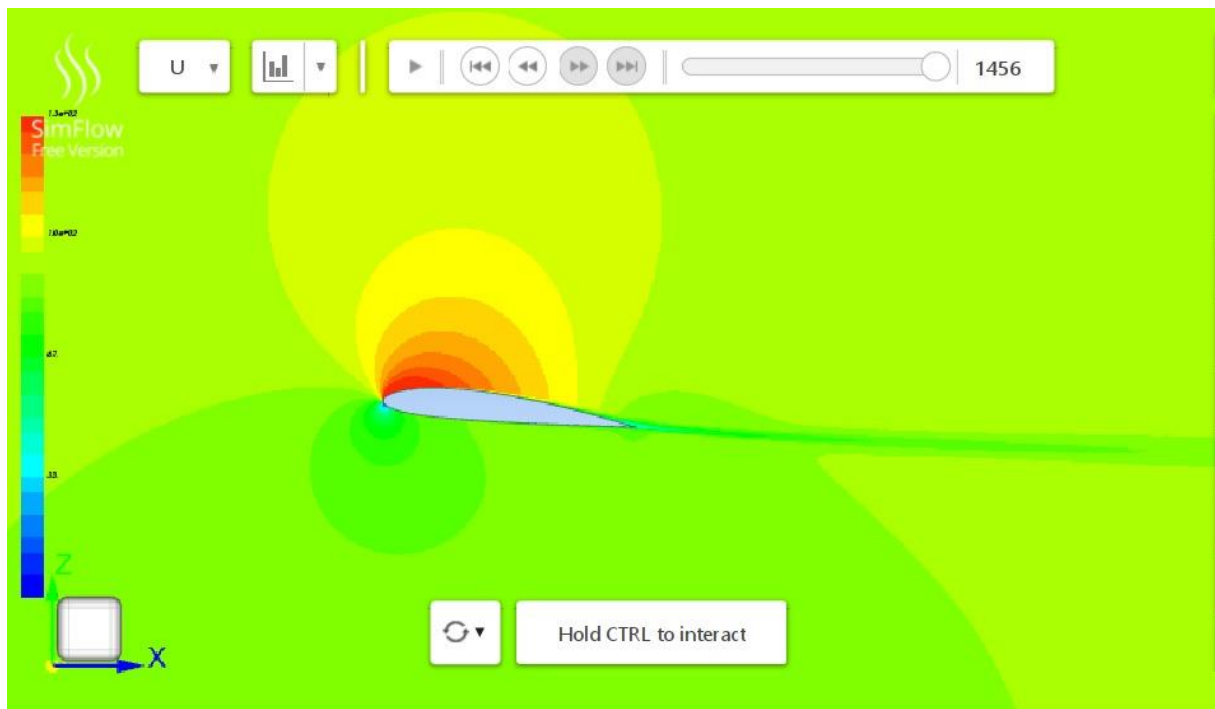


Figure 19: Velocity Contour

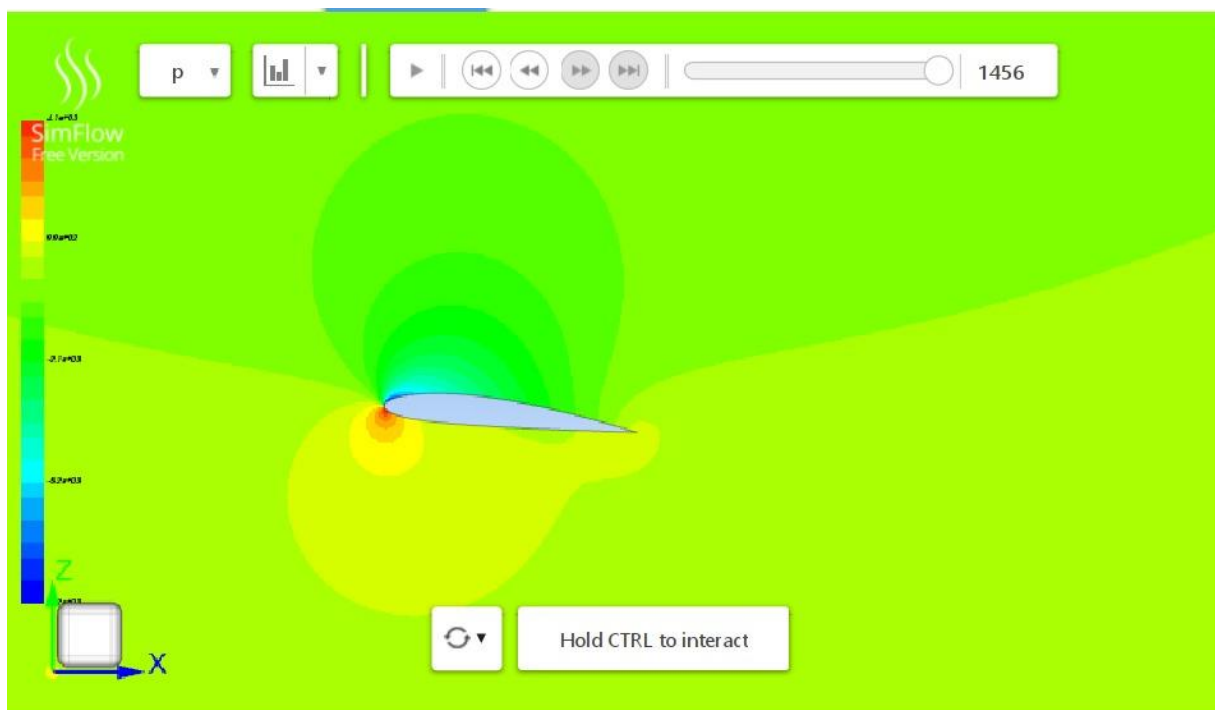


Figure 20: Pressure Contour

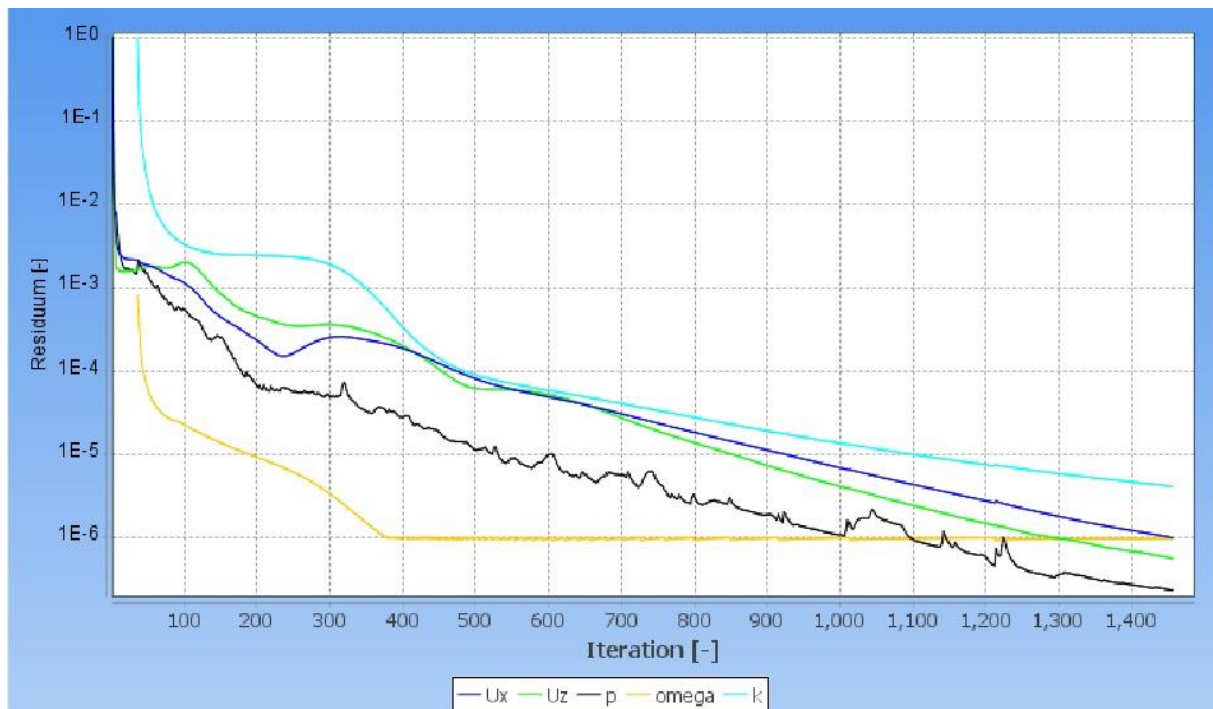


Figure 21: Residuals

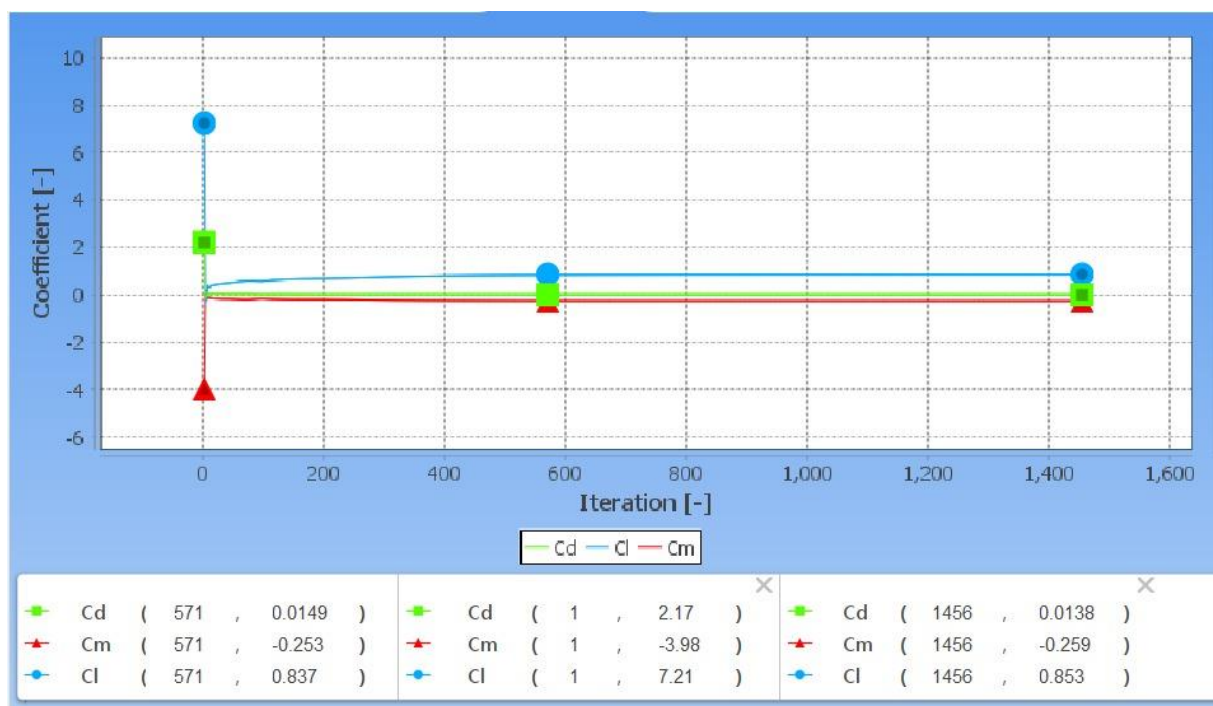


Figure 22:  $(C_d)$ ,  $(C_l)$ ,  $(C_m)$  values for 6-degree angle of attack

Results of 9 Degrees AoA:

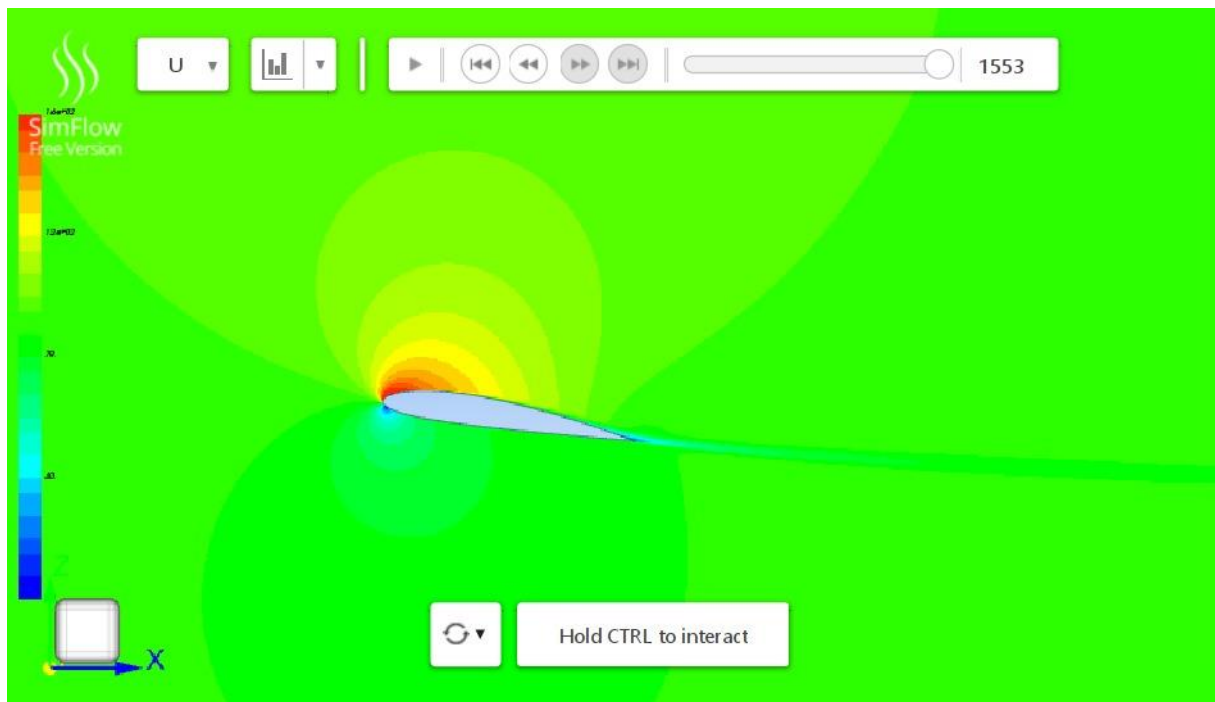


Figure 23: Velocity Contour

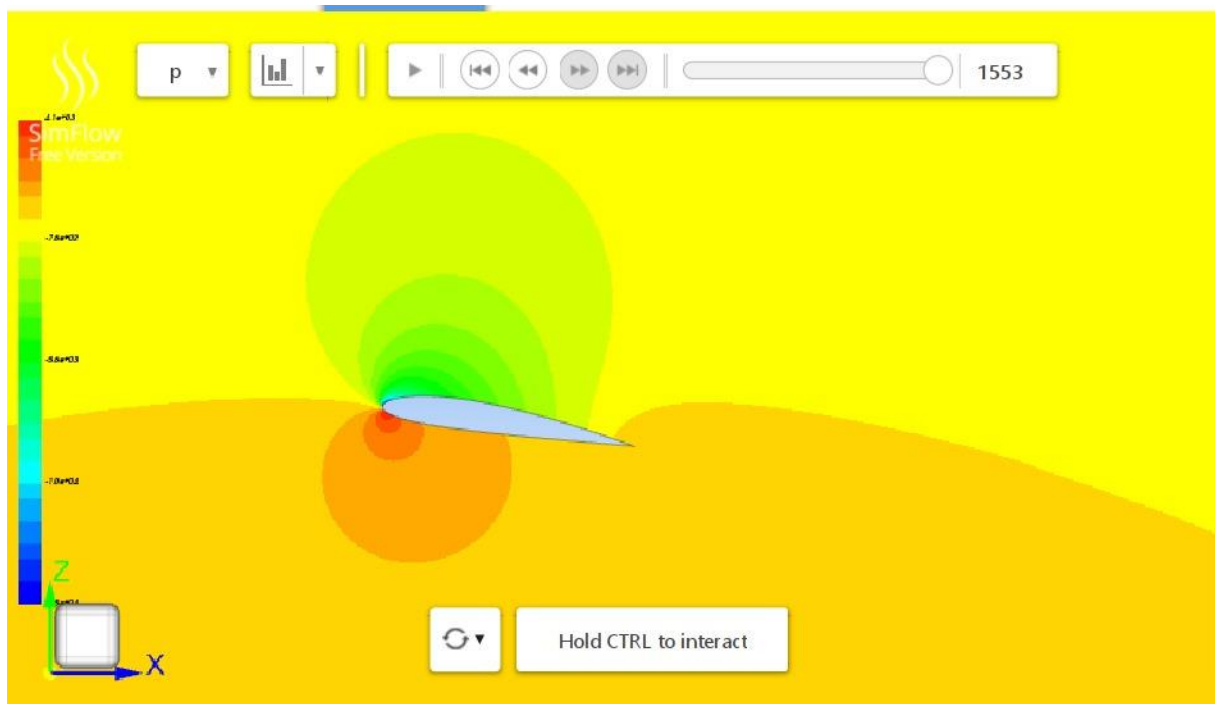


Figure 24: Pressure Contour

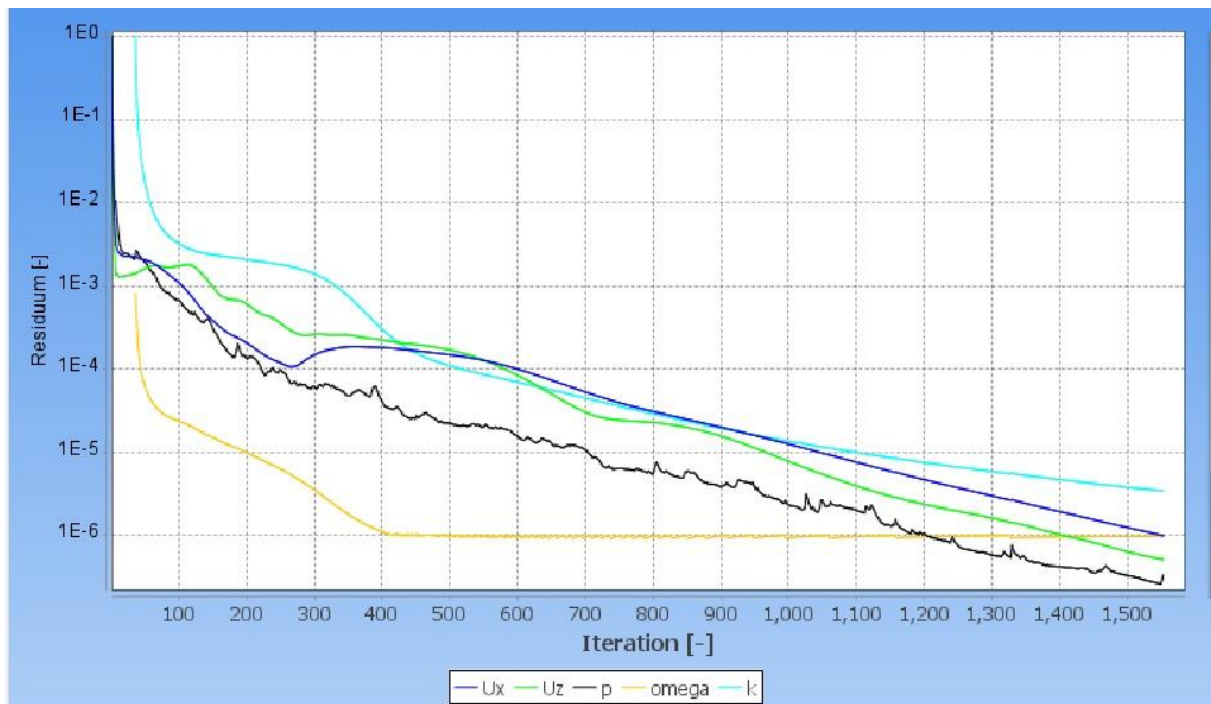


Figure 25: Residuals



Figure 26:  $(C_d)$ ,  $(C_l)$ ,  $(C_m)$  values for 9-degree angle of attack

## 15 Degrees AoA (Stall Point)

At a 15-degree angle of attack, the flow dynamics experience a critical breakdown, which is visually evident in the velocity contour map. The boundary layer can no longer overcome the severe adverse pressure gradient, causing massive flow separation that starts closer to the leading edge and extends over the entire upper surface. This creates a vast, low-velocity wake region (indicated by the large blue area trailing the airfoil). Consequently, the pressure distribution is severely disrupted, leading to a loss of suction on the upper surface. The aerodynamic coefficients reflect this phenomenon perfectly: the lift coefficient ( $C_L$ ) drops to 1.010, while the drag coefficient ( $C_D$ ) spikes dramatically to 0.0303. These visual and numerical results conclusively confirm that the NACA 2412 airfoil has entered a deep stall condition at 15 degrees.

## Results of Stall Point

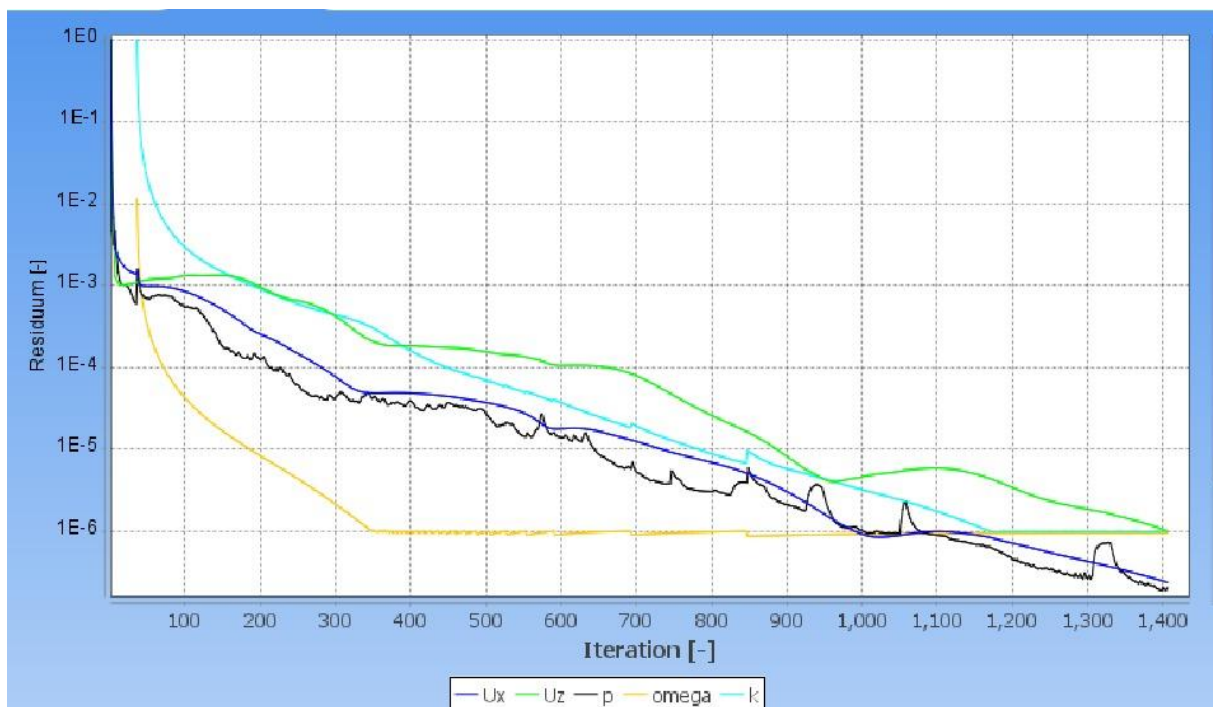


Figure 27: Residuals for 15-degree angle of attack

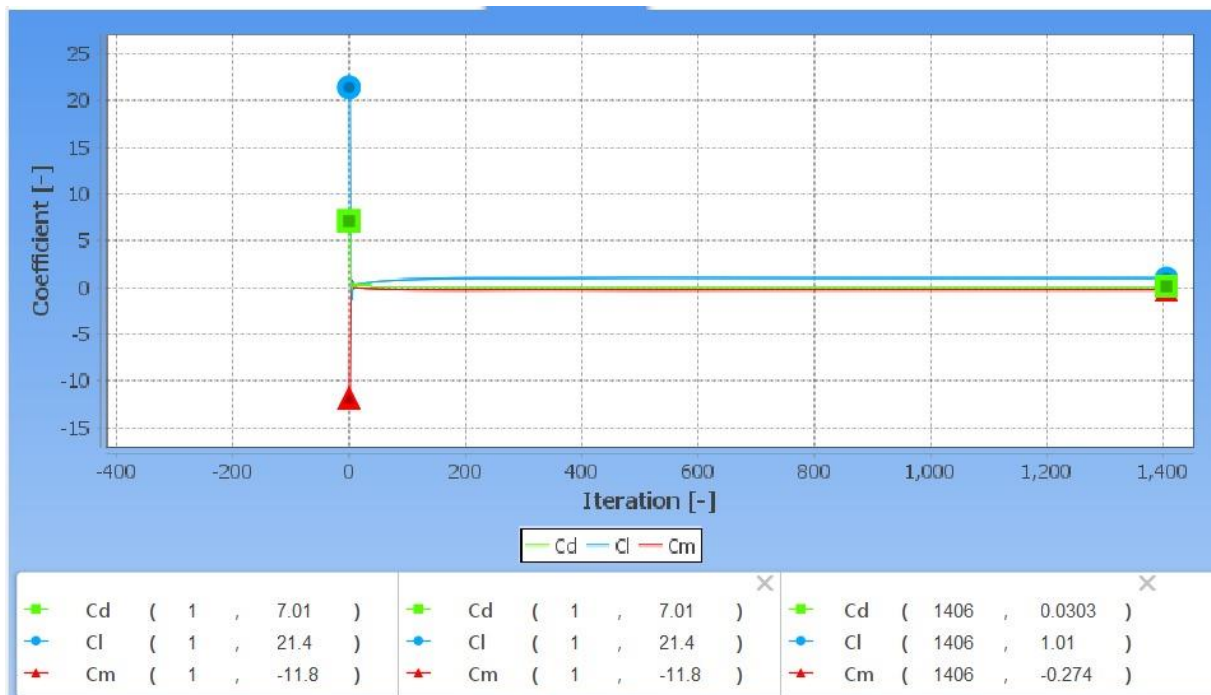


Figure 28: Cd, Cl, Cm values for 15-degree angle of attack

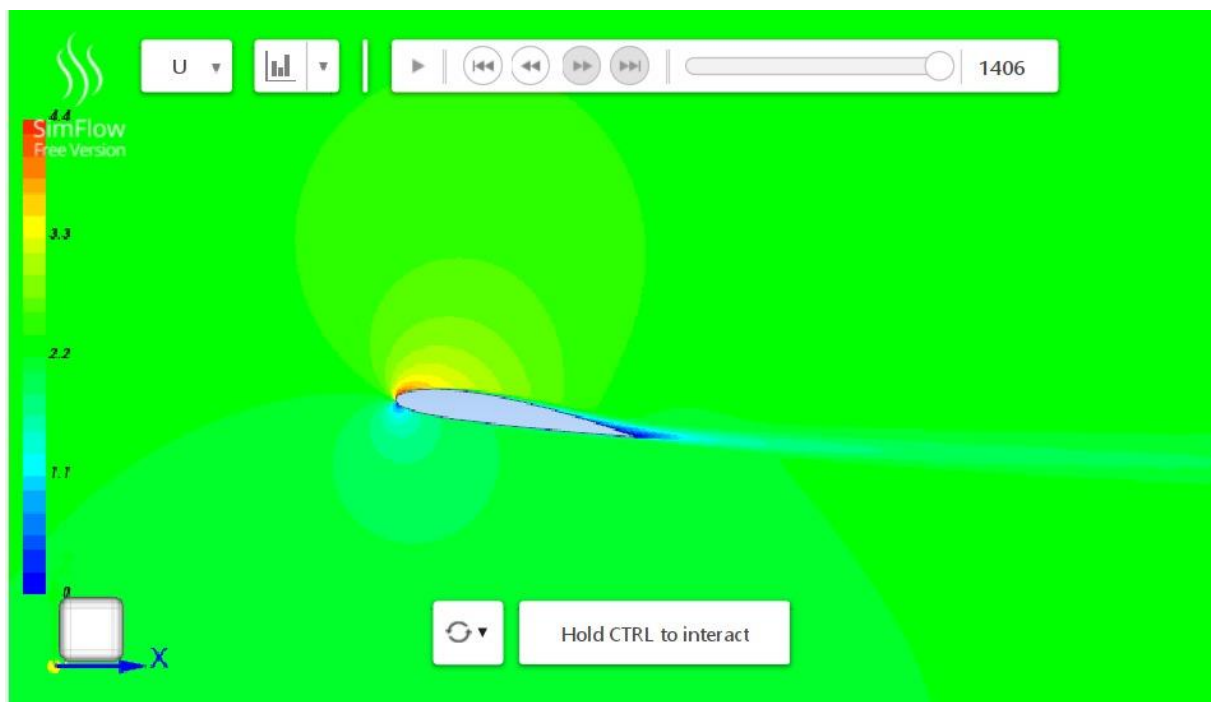


Figure 29: Velocity Contour

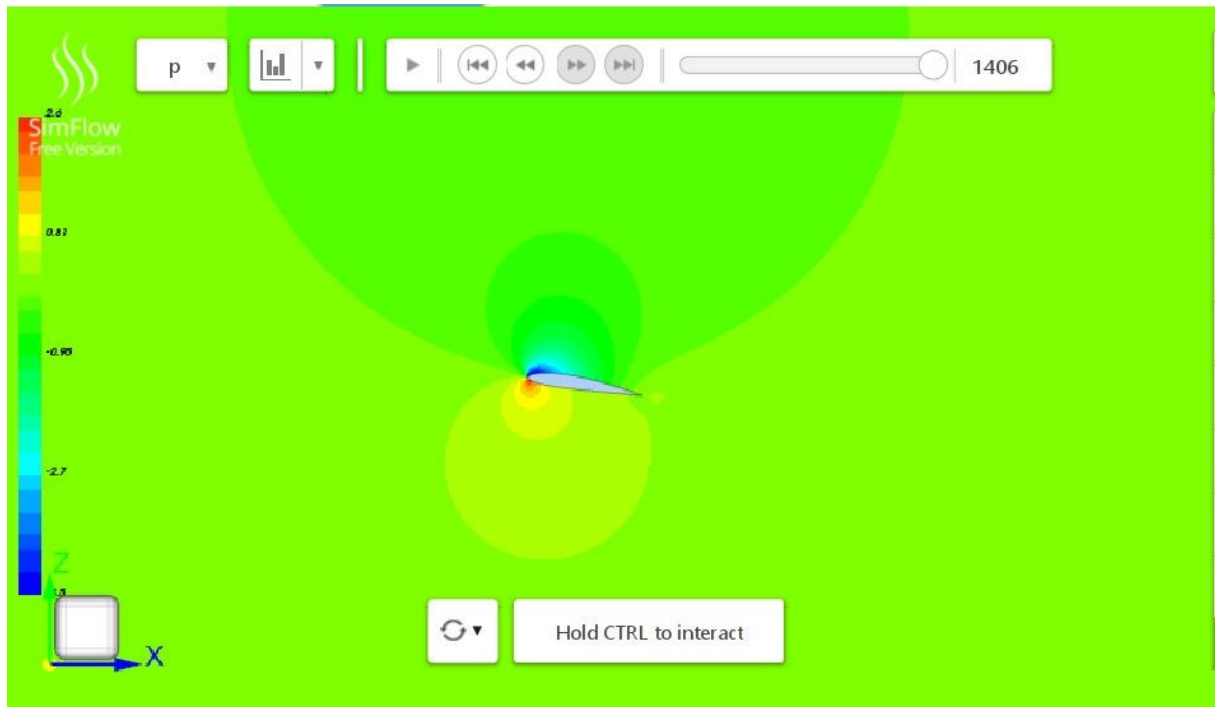


Figure 30: Pressure Contour

**Interpretation of Contours:** At low angles of attack ( $0^\circ$  and  $3^\circ$ ), the flow remains smoothly attached to the airfoil surface (attached flow). In the pressure contours, a high-pressure region forms on the lower surface and a distinct low-pressure region forms on the upper surface, generating the fundamental lift force. However, when the angle of attack reaches the stall point, a vast low-velocity wake region starting from the trailing edge and spreading over the upper surface is clearly visible in the velocity contours. This massive flow separation disrupts the pressure gradient that keeps the airfoil airborne, visually confirming the stall condition.

## 5. Parametric Analysis and Evaluation

### 5.1. Effect of Maximum Camber on Lift Coefficient (NACA 1412 vs 2412 vs 4412 vs 6412)

| EFFECT OF MAXIMUM CAMBER ON LIFT COEFFICIENT (AoA = 5°, Re = 1.0 × 10 <sup>5</sup> ) |                    |                              |               |                     |                       |                    |                    |                    |
|--------------------------------------------------------------------------------------|--------------------|------------------------------|---------------|---------------------|-----------------------|--------------------|--------------------|--------------------|
| Airfoil                                                                              | Maximum Camber (%) | Camber Position (% of chord) | Thickness (%) | Angle of Attack (°) | Reynolds Number       | C <sub>L</sub> (-) | C <sub>D</sub> (-) | C <sub>M</sub> (-) |
| NACA 0012                                                                            | 0                  | —                            | 12            | 5                   | 1.0 × 10 <sup>5</sup> | 0.548              | 0.0150             | 0.000              |
| NACA 1412                                                                            | 1                  | 4                            | 12            | 5                   | 1.0 × 10 <sup>5</sup> | 0.655              | 0.0162             | -0.025             |
| NACA 2412                                                                            | 2                  | 4                            | 12            | 5                   | 1.0 × 10 <sup>5</sup> | 0.767              | 0.0185             | -0.052             |
| NACA 4412                                                                            | 4                  | 4                            | 12            | 5                   | 1.0 × 10 <sup>5</sup> | 0.972              | 0.0210             | -0.105             |
| NACA 6412                                                                            | 6                  | 4                            | 12            | 5                   | 1.0 × 10 <sup>5</sup> | 1.150              | 0.0245             | -0.155             |

Figure 31: Aerodynamic coefficients for different maximum camber values (AoA = 5°, Re = 100,000)

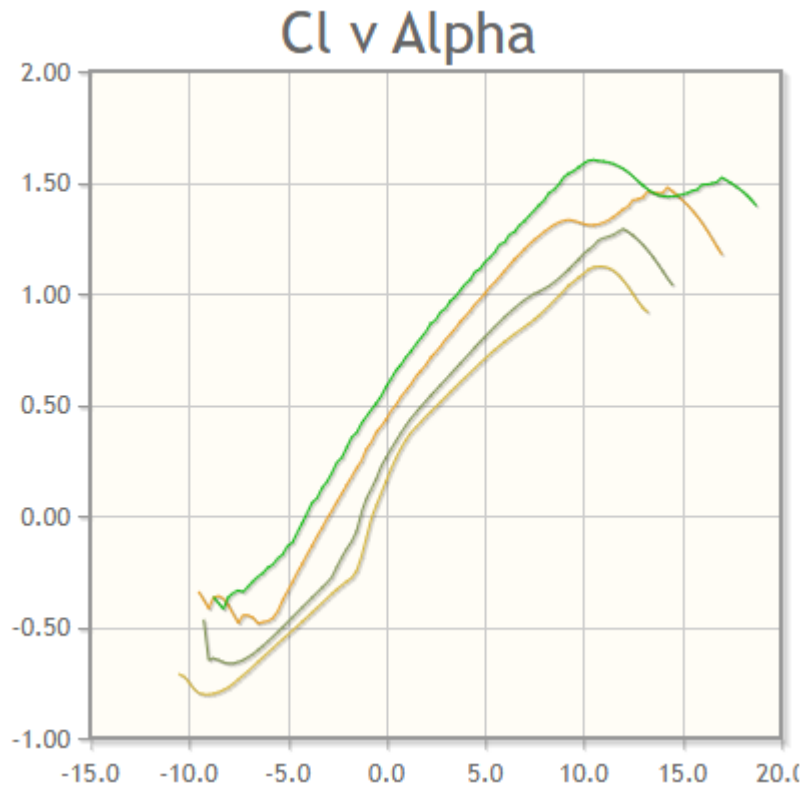


Figure 32: Effect of Maximum Camber on Lift Coefficient ( $Re = 100,000$ ) (Legend: Green line = NACA 6412, Orange line = NACA 4412, Dark Olive line = NACA 2412, Yellow line = NACA 1412)

By keeping the camber position, thickness, and Reynolds number ( $Re=100,000$ ) constant, the effect of increasing the maximum camber was analyzed by comparing four different profiles: NACA 1412, 2412, 4412, and 6412. As the maximum camber increases from 1% to 6%, the asymmetry between the upper and lower surfaces of the airfoil becomes more pronounced. This creates a stronger low-pressure region on the upper surface, resulting in a significant and direct increase in the lift coefficient ( $C_L$ ) at any given angle of attack. However, highly cambered airfoils (such as the NACA 6412) tend to experience adverse pressure gradients earlier, which can lead to premature flow separation and a lower stall angle compared to airfoils with less camber.

## 5.2. Effect of Position of Maximum Camber on Lift Coefficient

| Effect of Maximum Camber Position on Lift Coefficient |                 |                           |           |       |
|-------------------------------------------------------|-----------------|---------------------------|-----------|-------|
| Airfoil                                               | Max. Camber (%) | Camber Position (% chord) | AoA (deg) | Cl    |
| NACA 2112                                             | 2               | 10                        | 5         | 0.860 |
| NACA 2212                                             | 2               | 20                        | 5         | 0.830 |
| NACA 2312                                             | 2               | 30                        | 5         | 0.800 |
| NACA 2412                                             | 2               | 40                        | 5         | 0.767 |
| NACA 2512                                             | 2               | 50                        | 5         | 0.730 |

Figure 33: Lift coefficient values for different maximum camber positions (AoA = 5°, Re = 100,000)

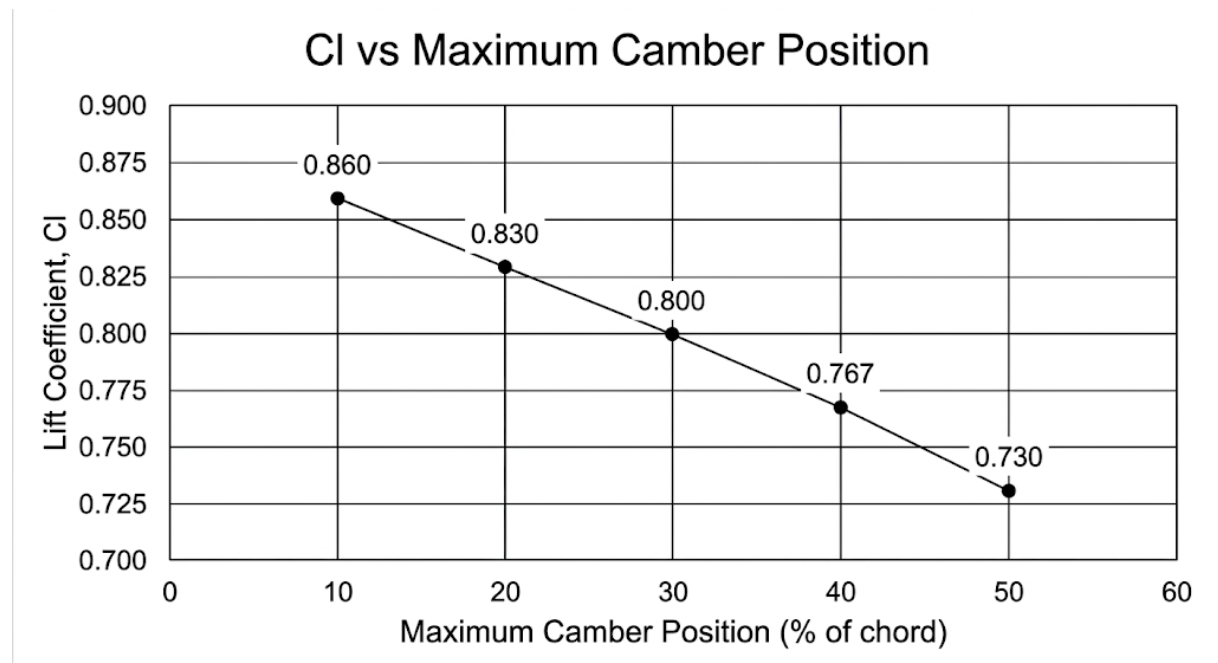


Figure 34: Lift coefficient ( $C_l$ ) vs Maximum Camber Position (% of chord)

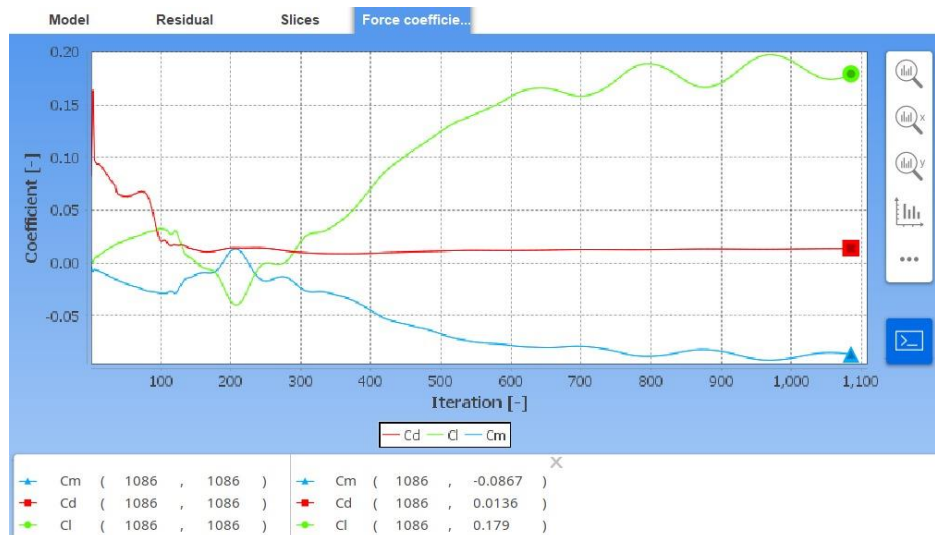
The position of the maximum camber significantly influences the aerodynamic performance of the airfoil. To isolate this effect, the maximum camber position was shifted from 10% to 50% of the chord (comparing NACA 2112, 2212, 2312, 2412, and 2512) while keeping the maximum camber (2%), thickness (12%), Reynolds number (Re=100,000), and angle of attack (AoA = 5°) constant.

As seen in the graph, moving the maximum camber position toward the trailing edge (from 10% to 50%) results in a linear decrease in the lift coefficient ( $C_l$ ). A forward camber position (e.g., NACA 2112) creates a stronger flow acceleration over the upper surface near the leading

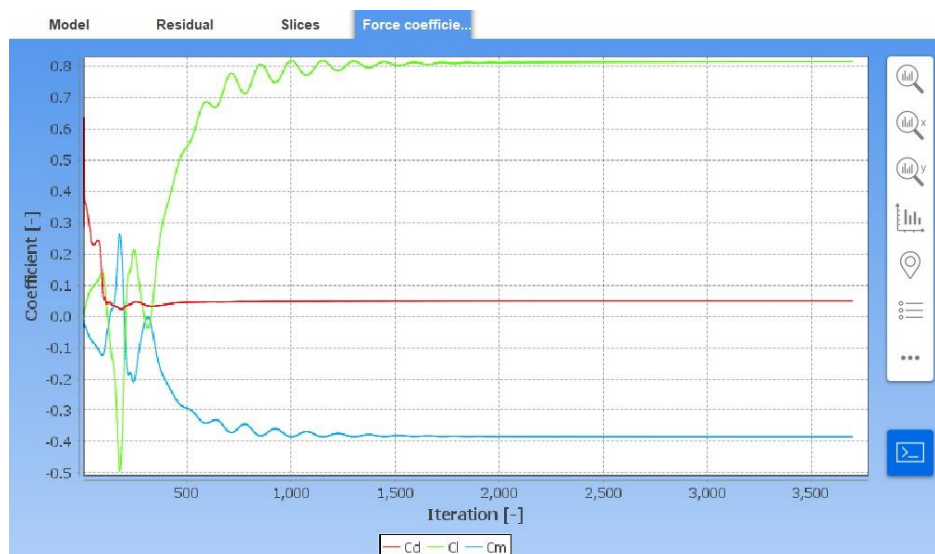
edge, generating a larger pressure difference and higher lift. Conversely, shifting the camber rearward weakens this pressure gradient near the front, thereby reducing the overall lift. Therefore, airfoils with a forward maximum camber position produce higher lift coefficients under these specific flow conditions.

### 5.3. Effect of Reynolds Number on Lift Coefficient

#### Velocity 5m/s



#### Velocity 10m/s



## Velocity 20 m/s



## Velocity 30 m/s



To evaluate the influence of the freestream velocity (Reynolds number) on the aerodynamic performance, the NACA 2412 airfoil was analyzed at a constant **angle of attack (AoA=5 °)** across varying velocities (5, 10, 20, and 30 m/s).

**Note on Raw Data and Mathematical Scaling:** During the parametric CFD simulations, the reference velocity ( $V_{ref}$ ) within the force coefficient monitors was kept constant at **10 m/s for all runs**. Since the solver calculates the dimensionless coefficients based on this constant reference dynamic pressure, the raw coefficients displayed in the solver monitors had to be analytically corrected for velocities other than 10 m/s. To obtain the true physical aerodynamic coefficients ( $C_L$  and  $C_D$ ), the raw values extracted from the software were multiplied by the scaling factor of  $(V_{ref} / V_{actual})^2$ . For instance, the raw values obtained from the 30 m/s analysis were multiplied by  $(10/30)^2$  to calculate the actual physical coefficients. The mathematically corrected and finalized values are presented in the table below.

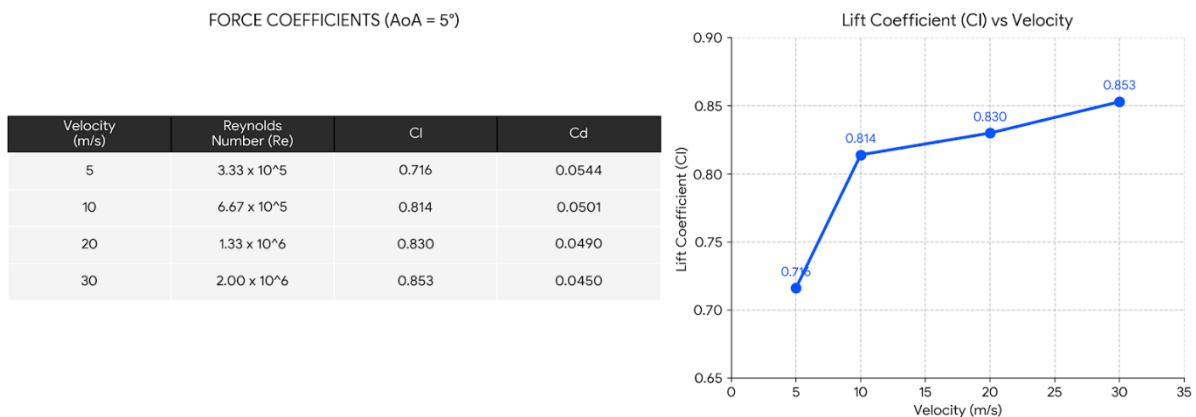


Figure 35: Effect of freestream velocity (Reynolds number) on aerodynamic coefficients for NACA 2412 airfoil (AoA=5 ° )

### Aerodynamic Evaluation:

As illustrated in the table, increasing the freestream velocity and consequently the Reynolds number positively impacts the overall aerodynamic characteristics of the airfoil. At a lower velocity of 5 m/s, the boundary layer possesses less kinetic energy, making it more susceptible to adverse pressure gradients and viscous effects. This condition results in a relatively lower lift coefficient ( $C_L=0.716$ ) and a higher drag coefficient ( $C_D=0.0544$ ).

However, as the freestream velocity is increased up to 30 m/s, the energized boundary layer becomes significantly more resistant to flow separation. This enhanced flow attachment allows the airfoil to achieve its maximum lift coefficient ( $C_L=0.853$ ) within the tested range, while simultaneously reducing viscous drag to its minimum ( $C_D=0.0450$ ). In conclusion, operating at higher Reynolds numbers successfully maximizes the aerodynamic efficiency ( $C_L/C_D$ ) of the NACA 2412 airfoil under steady flow conditions.

## 6. Conclusion

In this study, the aerodynamic characteristics of the NACA 2412 airfoil were comprehensively investigated using steady-state CFD simulations in SimFlow and verified with parametric XFOIL data. The primary objective was to analyze the quantitative and qualitative impacts of the angle of attack, maximum camber, camber position, and Reynolds number on overall aerodynamic performance.

The CFD simulation results demonstrated that increasing the angle of attack linearly increases the lift coefficient up to 9 degrees, where the airfoil achieves its peak aerodynamic efficiency ( $C_L/C_D$ ). However, at a high angle of attack of 15 degrees, a massive flow separation and a large wake region were observed on the upper surface, confirming a severe stall condition accompanied by a dramatic spike in the drag coefficient. Furthermore, the parametric analysis revealed that increasing the maximum camber is the most dominant geometric design factor for generating higher lift, while positioning the camber forward (closer to the leading edge) optimizes upper-surface velocity acceleration. Additionally, the velocity analysis indicated that operating at higher freestream velocities (up to 30 m/s) significantly improves overall aerodynamic efficiency by maximizing the lift coefficient and minimizing viscous drag.

In conclusion, the NACA 2412 airfoil exhibits stable, predictable, and highly efficient aerodynamic performance at low to moderate angles of attack, validating its widespread and fundamental selection in general aviation and UAV aerodynamic designs.